Stuff We Already Know

- Given a right-angled triangle with angle x, write down the definitions of the following:

$$\sin x = _____, \quad \cos x = _____$$

$$\tan x = _____, \quad \sec x = _____$$

$$\csc x = _____, \quad \cot x = _____$$

- Write down the following limits:

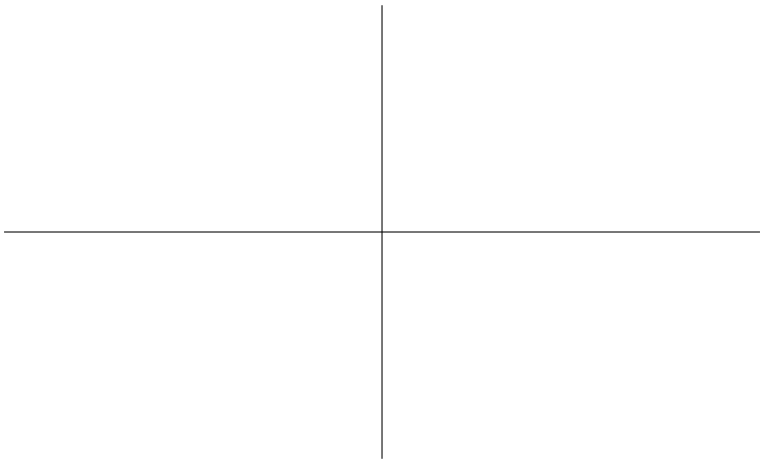
$$\lim_{x\to 0} \frac{\sin x}{x} = _____ , \qquad \lim_{x\to 0} \frac{\cos x - 1}{x} = _____.$$

- Write down the definition of the derivative for a function $f(x)$:

Derivatives of sin and cos

Questions

1. Graph the function, $g(x) = \frac{\sin(x+.001)-\sin(x)}{.001}$ on the axes below with domain $[-2\pi, 2\pi]$. What does this approximate?



2. What other function that we know does this graph look like? Graph it on top of your graph of $g(x)$.

3. Write down a formula (using a limit) that would give the derivative of $\sin x$:

$$\frac{d}{dx}(\sin x) = \lim _____.$$

4. Use the fact that $\sin(A+B) = \sin A \cos B + \cos A \sin B$ to rewrite the derivative of $\sin(x)$ as the sum of two different limits.

5. Use your two limits to find what $\frac{d}{dx} \sin x$ should be.

Question Use the same sorts of tricks to find $\frac{d}{dx} \cos x$:
(Note: $\cos(A+B) = \cos A \cos B - \sin A \sin B$.)

Other Trig Derivatives

Questions

1. Compute $\frac{d}{dx}\tan x$. (Hint: Quotient Rule)

2. Compute $\frac{d}{dx}\sec x$, $\frac{d}{dx}\csc x$, and $\frac{d}{dx}\cot x$. (Hint: for example, write $\sec x = \frac{1}{\cos x} = (\cos x)^{-1}$ and use the chain rule...)

The derivatives we found above are important. For future reference, write them below:

$\frac{d}{dx}\sin x =$ $\frac{d}{dx}\cos x =$

$\frac{d}{dx}\csc x =$ $\frac{d}{dx}\sec x =$

$\frac{d}{dx}\tan x =$ $\frac{d}{dx}\cot x =$

Questions

1. Find the equation of the tangent line to the graph of $f(x) = \tan x$ at $x = 1$.

2. A particle is moving along a straight line. Its position from its origin (in meters) at time t seconds is given by $s(t) = \sin^2 t$. Find its velocity and acceleration at time $t = 2$ seconds. At what times is the particle at rest? (Hint: The identity $\sin 2x = 2 \sin x \cos x$ may be useful.)

3. (a) Find the linear approximation to the curve $g(x) = \sec x$ at $x = \frac{5\pi}{6}$.

   (b) Use your answer above to estimate $\sec \frac{11\pi}{12}$.